



Spheretech AI Assistant

RAG Pipeline — API-Driven AI Assistant

FAISS

Groq LLaMA 3.3 70B

Streamlit

FastAPI

Docker

Problem & Solution

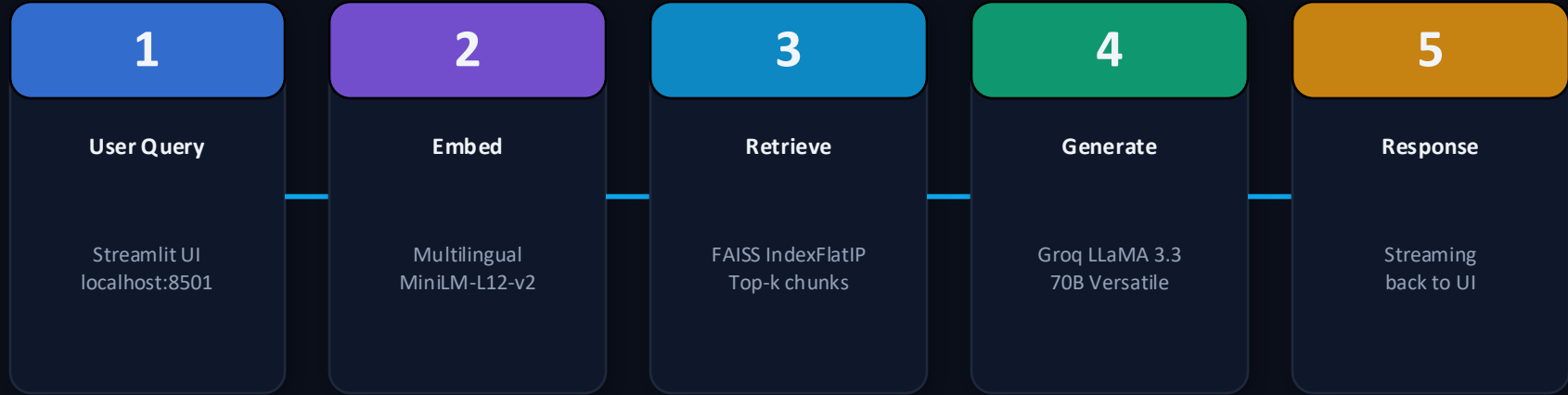
Problem

- Customers struggle to find fast answers about products and pricing
- Support team repeatedly answers the same questions
- Static FAQ pages cannot understand natural language queries
- Serving a multilingual user base is challenging

Solution

- Knowledge base with 50 Q&A chunks across 19 categories
- Semantic search retrieves the most relevant content
- Groq LLaMA 3.3 70B generates natural language answers
- Turkish-English cross-lingual support out of the box

System Architecture



FastAPI Backend (Bonus) `/health` `/categories` `/retrieve` `/ask` `/ask/stream` Swagger UI: `localhost:8000/docs`

Streamlit imports backend directly (Simple RAG) | Streamlit calls FastAPI via HTTP (Bonus mode)

Tech Stack

Embeddings

paraphrase-multilingual-MiniLM-L12-v2

- 50+ languages including Turkish
- Cross-lingual: TR query matches EN chunk
- 384-dim vectors, runs fully local

Vector Store

FAISS IndexFlatIP

- L2-normalized -> cosine similarity
- Exact search optimal for 50 chunks
- Category metadata for filtering

LLM

LLaMA 3.3 70B — Groq API

- GPT-4o level quality
- 394 tokens/sec, free tier
- temperature=0.2 for factual answers

Frontend

Streamlit

- Streaming token-by-token response
- Dark theme, session stats
- Deployed: streamlit.app

Backend API

FastAPI + Uvicorn

- Auto Swagger UI at /docs
- Pydantic validation, CORS
- Modern lifespan management

DevOps

Docker + Docker Compose

- Separate backend/frontend images
- Non-root user, health checks
- HF model pre-cached in image

Knowledge Base

50

Q&A Chunks

19

Categories

4

Edge Case Types

Categories

- Products (SphereShield, SphereSOC, ThreatRadar)
- Services (MDR, Pentest, Incident Response)
- Compliance (ISO 27001, GDPR, KVKK, SOC 2)
- Pricing & Technical Support
- Edge Case - Turkish | Negative Tone
- Edge Case - Cross Category | Competitor

Embedding Strategy

```
"Question: {q}\nAnswer: {a}"
```

Combining question + answer before embedding significantly improves retrieval accuracy.

Category stored as FAISS metadata, enabling category-level filtering.

Live Demo

AI ASSISTANT

Sphertech AI Assistant

Settings

top-k: 3

Filter: All

Streaming: ON

Queries: 2

Fiyatlariniz nedir? (What are your prices?)

Sphertech products are licensed on an annual subscription model. SphereShield starts at \$15 per endpoint per month with volume discounts for larger deployments...

Retrieved sources (score: 0.493 0.490 0.397)

Retrieval Pipeline

1

Query Embedding

```
get_query_embedding(query)
-> 384-dim vector
normalize_embeddings=True
```

2

FAISS Search

```
index.search(query_vec,
top_k)
IndexFlatIP
Inner Product = Cosine
Sim
```

3

Score Filter

```
score_threshold=0.0
category_filter=None
Top-k ranked results
```

4

Context Format

```
"[1] Category: Products
Q: SphereShield?
A: ..."
```

Turkish 'Fiyatlariniz nedir' matched 'Pricing and Licensing' chunk with 0.493 cosine similarity -- cross-lingual retrieval successful!

Bonus Features

FastAPI Backend

Completed

5 endpoints: /health /categories /retrieve /ask /ask/stream
Pydantic schemas, CORS, modern lifespan management (not deprecated on_event)
Swagger UI: localhost:8000/docs

Streaming Response

Completed

generate_answer_stream() yields SSE token-by-token
Word-by-word rendering in Streamlit UI
FastAPI /ask/stream endpoint also available

Evaluation Methodology

Proposed

Hit Rate @k -- is the correct chunk in the top-k results?
MRR (Mean Reciprocal Rank), ROUGE-L, BERTScore, Faithfulness
Working code example included in README

Design Decisions

Why multilingual MiniLM?

Knowledge base contains Turkish edge cases. 50+ languages -- a Turkish query can match an English chunk seamlessly.

Why IndexFlatIP?

Embeddings are L2-normalized so inner product equals cosine similarity. Exact search optimal for 50 chunks. IndexIVFFlat preferred at 10K+.

Why temperature=0.2?

Low temperature produces factual and consistent answers. Hallucinations in a customer assistant erode user trust.

Why Stateless RAG?

Case study requests Simple RAG. Chat history displayed for UX only; only the latest question is sent to LLM. LangChain memory can be added next phase.



Project Summary

50 chunks | 19 categories | Turkish edge cases

Multilingual RAG + cross-lingual retrieval

LLaMA 3.3 70B | Streaming | FastAPI | Docker

Live: spheretech-rag.streamlit.app | All bonuses completed

Next Steps

- Multi-turn memory via LangChain
- Evaluation pipeline (Hit Rate, ROUGE-L)
- Document ingestion API for new content
- Streamlit Cloud + CI/CD pipeline